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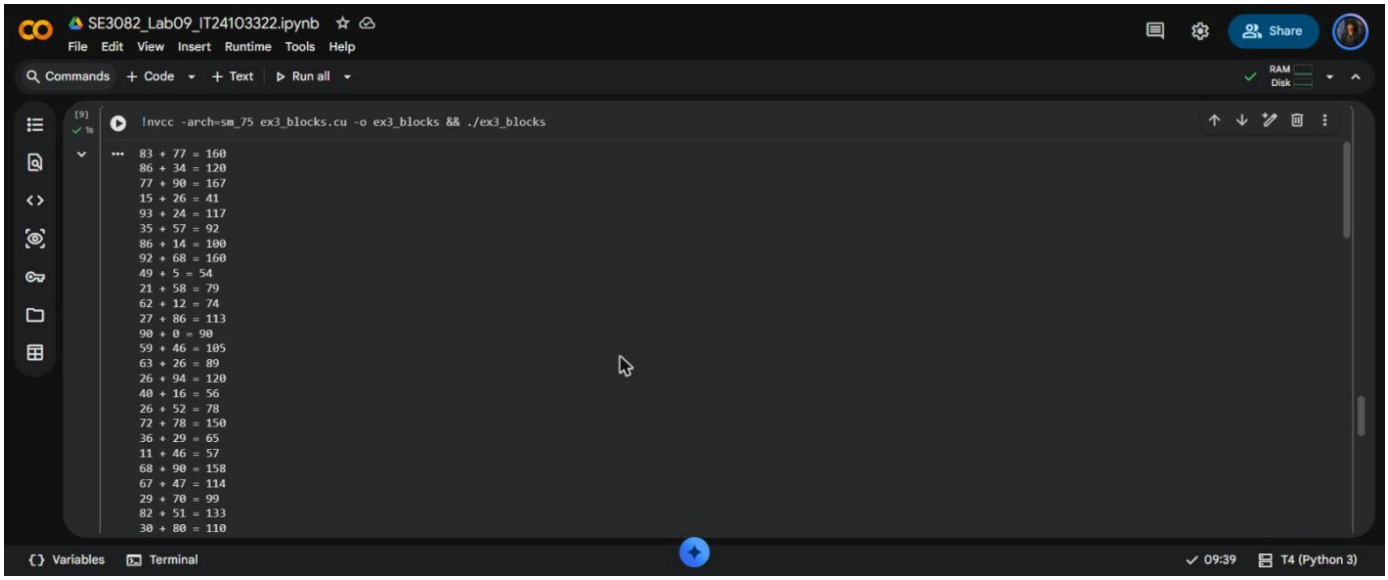
BSc (Hons) in Computer Science
Year 3

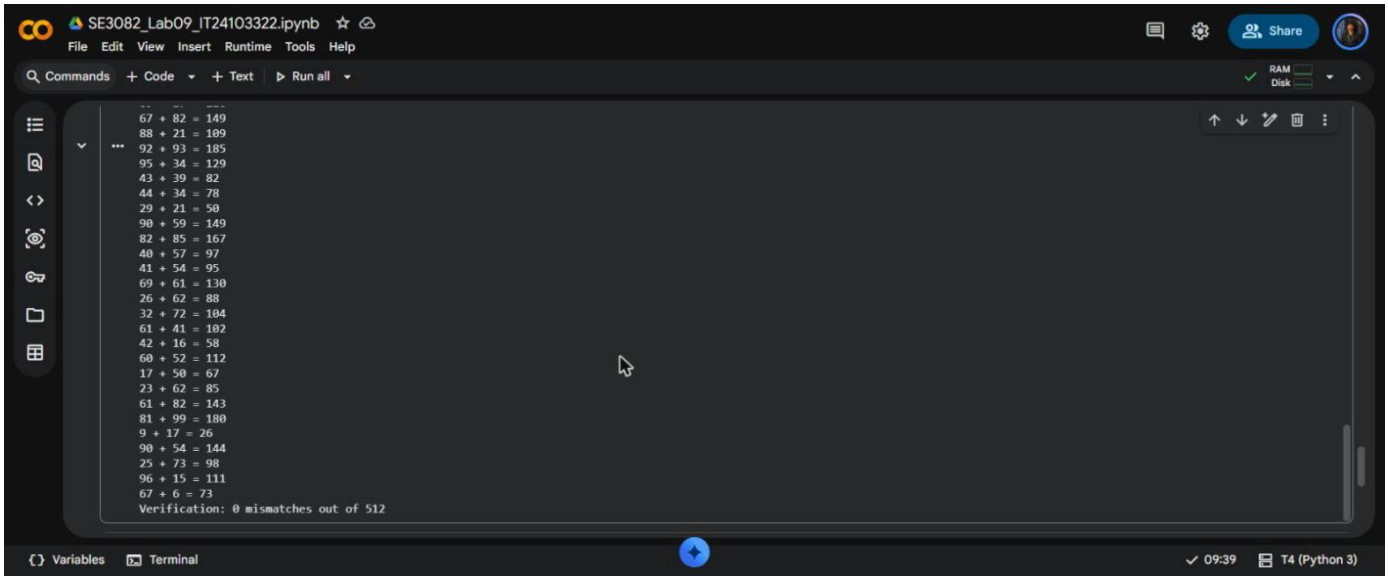
Lab Sheets

SE3082 – Parallel Computing **Semster 1, 2026**

IT24103322

Exercise 3





Two vectors of 512 integers (rand() % 100) are added with add<<<N, 1>>>, so the kernel runs in 512 blocks with one thread each and every block uses blockIdx.x to pick its element: c[blockIdx.x] = a[blockIdx.x] + b[blockIdx.x].

The first lines are 83 + 77 = 160, 86 + 34 = 120 and 77 + 90 = 167, as in the lab sheet, and the last line is Verification: 0 mismatches out of 512, so every GPU result matches the CPU.

1. **How many blocks and how many threads in total does add<<<N, 1>>> launch? What does a warp of 32 mean for a block that contains only one thread?**

   It launches 512 blocks and 512 threads in total (512 × 1). The GPU runs threads in warps of 32, and a warp never contains threads from two different blocks. A block with one thread still takes a whole warp, so 1 lane does the work and the other 31 lanes are idle: only 1/32 of the GPU's lanes are used, which is why one thread per block is very inefficient.